those performance-critical parts of the application where there's the potential for measurable gain. Of course, if the potential for performance gains is only secondary to clarity of design or separation of concerns, it may still be worth using a multithreaded design.

Assuming that you've decided you *do* want to use concurrency in your application, whether for performance, separation of concerns, or because it's "multithreading Monday," what does that mean for C++ programmers?

1.3. Concurrency and multithreading in C++

Standardized support for concurrency through multithreading is a new thing for C++. It's only with the upcoming C++11 Standard that you'll be able to write multithreaded code without resorting to platform-specific extensions. In order to understand the rationale behind lots of the decisions in the new Standard C++ Thread Library, it's important to understand the history.

1.3.1. History of multithreading in C++

The 1998 C++ Standard doesn't acknowledge the existence of threads, and the operational effects of the various language elements are written in terms of a sequential abstract machine. Not only that, but the memory model isn't formally defined, so you can't write multithreaded applications without compiler-specific extensions to the 1998 C++ Standard.

Of course, compiler vendors are free to add extensions to the language, and the prevalence of C APIs for multithreading—such as those in the POSIX C standard and the Microsoft Windows API—has led many C++ compiler vendors to support multithreading with various platform-specific extensions. This compiler support is generally limited to allowing the use of the corresponding C API for the platform and ensuring that the C++ Runtime Library (such as the code for the exception-handling mechanism) works in the presence of multiple threads. Although very few compiler vendors have provided a formal multithreading-aware memory model, the actual behavior of the compilers and processors has been sufficiently good that a large number of multithreaded C++ programs have been written.